Samir Panja

Post Doctoral Fellow · Statistics and Mathematics Unit

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Education

Indian Institute of Technology Bombay — *Mumbai, India*
PhD Mathematics — *2017 - 2023*

Indian Institute of Technology Kanpur — *Kanpur, India*
Master of Science in Mathematics — *2014 - 2016*

Vidyasagar University (Kharagpur College) — *Midnapore, West Bengal, India*
Bachelor of Science in Mathematics Honours — *2011 - 2014*

Academic appointments

January, 2024 - Present	Post Doctoral Fellow Statistics and Mathematics Unit Indian Statistical Institute Bangalore Centre, Bangalore, India **Mentor:** Prof. Jaydeb Sarkar
August - December, 2023	Institute Post Doctoral Fellow Department of Mathematics Indian Institute of Technology Bombay, Mumbai, India **Mentor:** Prof. Bata Krishna Das
April - June, 2023	Research Associate Department of Mathematics Indian Institute of Technology Bombay, Mumbai, India **Mentor:** Prof. Bata Krishna Das

Area of specialization

Mathematics: Functional Analysis, Operator Theory, Complex Analysis, Function Theory

Research focuses: *Working on the interface of operator theory, functional analysis, complex analysis, and function theory. More precisely on dilation and commutant lifting, reproducing kernel Hilbert spaces, complete Nevanlinna–Pick spaces, de Branges–Rovnyak spaces, Toeplitz and Hankel operators, and noncommutative function theory. Also, interested in developing the theoretical foundations of machine learning and exploring their applications.*

Publications, preprints, and book chapters

- Sneha B., N, Bala, **S. Panja**, and J. Sarkar, *Lifting and invariant subspaces of Hankel operators*, Publicacions Matemàtiques 70 (2026), 379 - 396.
- B.K. Das and **S. Panja**, *Isometric dilation and Sarason's commutant lifting theorem in several variables*, Canadian Journal of Mathematics 78 (2026), 1047 - 1079.
- H. Ahmed, B.K. Das, and **S. Panja**, *de Branges-Rovnyak spaces which are complete Nevanlinna-Pick spaces*, Journal of Geometric Analysis 35 (2025), no. 2, Paper No. 61.
- **S. Panja**, *Factorization of Toeplitz operators*, New York Journal of Mathematics 29 (2023), 848 - 873.
- *M. Bhattacharjee, B.K. Das, R. Debnath, and **S. Panja**,* Dilations and operator models of $\mathcal{W}$-hypercontractions*, Complex Anal. Oper. Theory 17 (2023), no. 2, Paper No. 22.*
- S. Barik and **S. Panja**, *Noncommutative commutant lifting, interpolation, and shift-invariant subspaces*, arXiv:2609.11410.
- S. Barik, S. Jain, and **S. Panja**, *Doubly commuting invariant and co-invariant subspaces*, arXiv:2610.18403.
- **S. Panja**, *Toeplitz operators with bounded harmonic symbols on weighted Bergman spaces are the fixed points of a normal unital completely positive map*, a preprint.
- S. Barik, S. Jain, **S. Panja**, and J. Sarkar, *Range spaces in $H^2(\mathbb{D}^n)$: backward shift invariance, characterization and Gleason's problem*, a preprint.

Book chapters:

- M. Bhattacharjee, B.K. Das, R. Debnath and **S. Panja**, *Dilations and operator models of $\mathcal{W}$-hypercontractions*, Multivariable operator theory, Birkhäuser/Springer, Cham (2023), pp - 119 - 154.

Awards and recognitions

2023	**Post Doctoral Fellowship award**, National Board for Higher Mathematics (NBHM)
2017	**Ph. D. Scholarship**, National Board for Higher Mathematics (NBHM)
2017	**Ph. D. Scholarship**, Council of Scientific and Industrial Research (CSIR) NET
2017	**Eligibility for Lectureship**, Council of Scientific and Industrial Research (CSIR) LS-NET
2017	**Graduate Aptitude Test in Engineering (GATE)**, IITs and IISc
2015 - 2016	**Merit-cum-Means (MCM) Scholarship**, IIT Kanpur
2014	**Joint Admission Test for Masters (JAM)**, IITs and IIsc
2012 - 2014	**Merit-cum-Means (MCM) Scholarship**, Government of West Bengal

Teaching experience

Instructor, ISI Bangalore Centre Autumn 2024
Function Spaces, B.Math. 3rd year (Designed and taught the course)

Teaching Assistant, ISI Bangalore Centre Spring 2026
Functional Analysis, M.Math 1st year.

Teaching Assistant, IIT Bombay

Semester	Course	Student cohort
Spring 2018	Introduction to Numerical Analysis	B.Tech, 2nd year
Autumn 2018	Calculus	B.Tech, 1st year
Spring 2019	Ordinary Differential Equations	B.Tech, 1st year
Autumn 2019	Linear Algebra	B.Tech, 1st year
Spring 2020	Introduction to Numerical Analysis	B.Tech, 2nd year
Autumn 2020	Real Analysis	M.Sc., 1st year
Spring 2023	Calculus	B.Tech, 1st year

Teaching Assistant, Advanced Instructional School June 2023
BITS Pilani, Goa Campus: *Function Spaces; Operator Theory and Operator Algebras*.

Conference Talks

January, 2025 On **"Which de Branges-Rovnyak Spaces have Complete Nevanlinna-Pick Property?"**, PDF-RS Annual Symposium 2025, Indian Statistical Institute Bangalore Center, India

March, 2024 On **"Isometric Dilation and Commutant Lifting of Operator Tuples with Defect Zero"**, Conference on National Conference on Operator Theory and Function Spaces, Shiv Nadar University, India

February, 2023 On **"Isometric Dilation and Sarason's Commutant Lifting Theorem in Several Variables"**, Conference on Functional Analysis and Related Topics, Department of Mathematics, IIT Bombay, India

September 2022 On **"Dilations and Operator Models of $\mathcal{W}$-hypercontractions"**, International Workshop on Operator Theory and its Applications (IWOTA) at University of Agriculture in Krakow, Poland

Academic service and professional development

Conference organization: Conference on Functional Analysis and Related Topics, IIT Bombay, February 2023.

Conferences and workshops attended:

- Noncommutative Analysis and Probability, IISc Bangalore (January 2026);
- Operator Algebra and Related Topics, IIT Bombay (February 2025);
- PDF-RS Annual Symposium, ISI Bangalore (January 2025);
- Noncommutative Mathematics and Applications, ISI Bangalore (October 2024);
- National Conference on Operator Theory and Function Spaces, Shiv Nadar University (March 2024);
- Functional Analysis and Related Topics, IIT Bombay (February 2023);
- IWOTA, Krakow (September 2022);
- Fields Institute Focus Program on Analytic Function Spaces and Their Applications, online (July–December 2021);

- ATM School on Operator Theory, IIT Kanpur (December 2019);
- Functional Analysis conference, IIT Bombay (October 2019).

Technical skills and languages

Programming: Lean, Python, C++. **Tools:** LaTeX, Excel, Word, HTML, CSS.
Languages: English, Bengali, Hindi.

Contact

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Indian Statistical Institute Bangalore Centre
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Mobile No.: +91 8250645863

Referees

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Prof. Jaydeb Sarkar — Statistics and Mathematics Unit, ISI Bangalore.
jaydeb@gmail.com Tel.: +91 80 2698 5479.